

Agenda

- 1) local + Remote + Default Gateway
- 2) Subnet
- 3) VLSM
- 4) IPV₆ Addresses

Common CIDR Reference

CIDR	Subnet Mask	Usable Hosts	Common Use
/8	255.0.0.0	16,777,214	Large ISPs
/16	255.255.0.0	65,534	Large enterprises
/24	255.255.255.0	254	Home, office networks
/25	255.255.255.128	126	Medium subnets
/26	255.255.255.192	62	Small subnets
/27	255.255.255.224	30	Small offices
/28	255.255.255.240	14	Very small subnets
/30	255.255.255.252	2	Point-to-point links
/32	255.255.255.255	0 (single host)	Host routes

255.255.224.0 /23

192.168.4.135

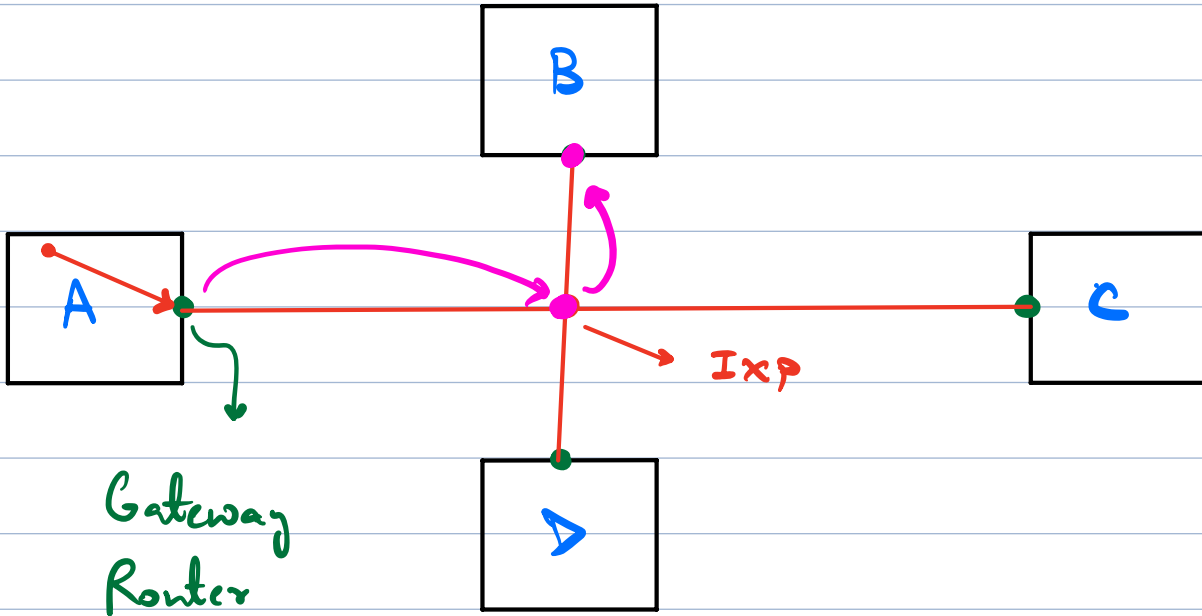
23 / (9) →

(2⁹) - 2

$$512 - 2 = \underline{\underline{510}}$$

Gateway $\Rightarrow$ 192.168.1.1

X $\Rightarrow$ IP



Subnetwork

Scaler $\Rightarrow$ 144.255.1.36 / 24

Network $\Rightarrow$ 144.255.1.0 / 24

Unique IP $\Rightarrow$ 2^8

Usable Hosts = $2^8 - 2$

$(2^8/2^2)$

144. 255. 1. 0 /24

SST

SSB

Academy

AI Labs

SST

144. 255. 1. 00 00000000

Network $\Rightarrow$ 144. 255. 1. 0 /26

Broadcast $\Rightarrow$ 144. 255. 1. 63

Host bits $\Rightarrow$ 144. 255. 1. 1 to 144. 255. 1. 62

SSB

144. 255. 1. 01 00000000

Network $\Rightarrow$ 144. 255. 1. 64 /26

Broadcast $\Rightarrow$ 144. 255. 1. 127

Host IPs $\Rightarrow$ 144. 255. 1. 65 to 144. 255. 1. 126

Academy

144. 255. 1. 10 00000000

N $\Rightarrow$ 144. 255. 1. 128 /26

B $\Rightarrow$ 144. 255. 1. 191

H $\Rightarrow$.129 to .190

AI Labs

144. 255. 1. 1 1 0 0 0 0 0 0

N $\Rightarrow$ _ . _ . _ . 192 /26

B $\Rightarrow$ _ . _ . _ . 255

H $\Rightarrow$. 193 to . 254

Steps

- 1) Determine how many hosts you need.
- 2) Find smallest $2^N \geq (\text{hosts} + 2)$
- 3) Host bits = 2^N
Prefix = $32 - N$
- 4) Subnet size (Block) = 2^N
- 5) Usable hosts = $2^N - 2$

Quick Reference: Host Count $\rightarrow$ Prefix

Hosts Needed	Min 2^N	Prefix	Usable Hosts
2	4	/30	2
3-6	8	/29	6
7-14	16	/28	14
15-30	32	/27	30
31-62	64	/26	62
63-126	128	/25	126
127-254	256	/24	254

Q Need 50 hosts per subnet within
 $10.0.0.0 / 24$

Org $\Rightarrow 192.168.10.0 / 24$

Eng $\Rightarrow 60$

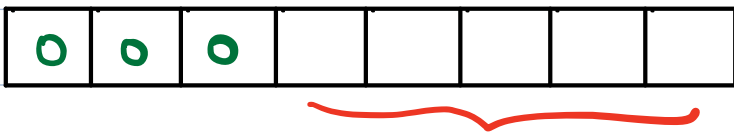
Marketing $\Rightarrow 30$

HR $\Rightarrow 10$

Management $\Rightarrow 5$

VLSM $\Rightarrow$ Variable Length Subnet Mask

Golden Rule $\Rightarrow$ Allocate largest subnet first, then work downwards.



Marketing $\Rightarrow 30 \Rightarrow /27$

000 00000 $\Rightarrow 0$

000 11111 $\Rightarrow 31$

Engineering



00 000000 $\Rightarrow 0$

00 111111 $\Rightarrow 63$

00 100000 32 -

00 111111 63

01 000000 64 -

01 011111 95

01 000000 64

01 111111 127

Org $\Rightarrow 192.168.10.0/24$

Eng $\Rightarrow 60 \Rightarrow /26$

Marketing $\Rightarrow 30$

HR $\Rightarrow 10$

Management $\Rightarrow 5$

Eng $\Rightarrow 60 \text{ Hosts} \Rightarrow /26$

Network $\Rightarrow$ 192.168.10.0 /26

0 0 0 0 0 0 0 0

Broadcast $\Rightarrow$ 192.168.10.63

0 0 1 1 1 1 1 1

Hosts $\Rightarrow$.1 to .62

Range left $\Rightarrow$.64 to .255

$\Rightarrow$ Marketing $\Rightarrow$ 30 $\Rightarrow$

Network $\Rightarrow$ 192.168.10.64 /27

0 1 0 0 0 0 0 0

Broadcast $\Rightarrow$ 192.168.10.95

0 1 0 1 1 1 1 1

Hosts. $\Rightarrow$.65 to .94

HR $\Rightarrow$ 10 $\Rightarrow$ /28

192.168.10.96 / 28

0 1 1 0 0 0 0 0

Broadcast $\Rightarrow$ 192.168.10.111

0 1 1 0 1 1 1 1

Hosts .97 to .110

Merge $\Rightarrow 5 + 2 = 7 \Rightarrow 8 \Rightarrow /29$

Network $\Rightarrow$ 192.168.10.112 / 29

Broadcast $\Rightarrow$ 192.168.10.119

Hosts $\Rightarrow$ 192.168.10.113 to 118

NAT (Network Address Translation)

Range	CIDR	Size	Common Use
10.0.0.0 – 10.255.255.255	10.0.0.0/8	16.7M addresses	Large corps, cloud VPCs
172.16.0.0 – 172.31.255.255	172.16.0.0/12	1.04M addresses	Medium networks
192.168.0.0 – 192.168.255.255	192.168.0.0/16	65,536 addresses	Home routers, offices

NAT translation table

(private IP : private port) → (public IP : translated port)

Public IP: 203.0.113.5

192.168.1.5 : 50001 → 142.250.115.46 : 80



203.0.113.5 : 40001 → 142.250.115.46 : 80

192.168.1.6 : 50001 → 142.250.115.46 : 80



203.0.113.5 : 40002 → 142.250.115.46 : 80

NAT Variants

SNAT → Outgoing ⇒ Only replaces IP not port

DNAT → Incoming ⇒ Port + IP

PAT (Port Address Translation)

⇒ Many devices share same IP

Port + IP
